Macro Dashboard Diagnostics Report

Prepared on February 1, 2026

Last edit on Monday, March 2, 2026

By: Chris Jose Castaneda

Table of Contents

Introduction **1**

Broad Macro Environment **1**

Interest Rates and Policy Conditions **2**

Inflation and Inflation Persistence **3**

Labor Market Conditions **4**

Market Volatility and Regime Comparisons **5**

Relationships and Stability Across Time **6**

Yield Curve and Forward Risk Signals **7**

Conclusion **8**

References ..... **A**

Appendix ..... **B**

Table 1: ..... B

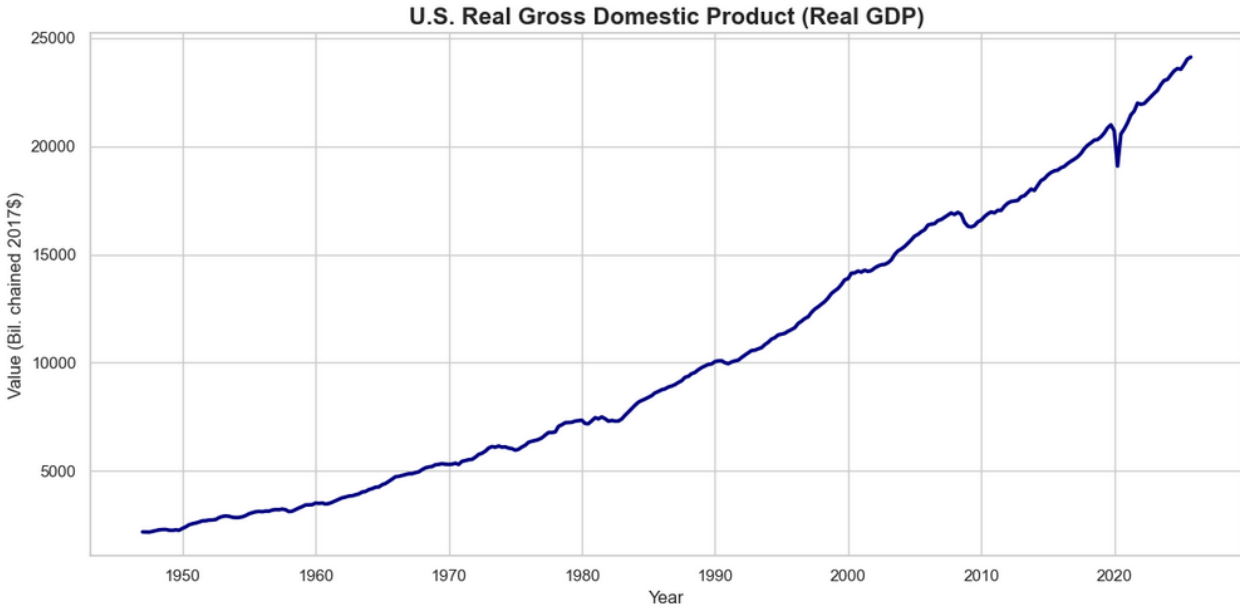
Introduction

This report evaluates the current U.S. macroeconomic environment using a dashboard constructed from Federal Reserve Economic Data (FRED). The analysis integrates key indicators across growth, inflation, labor market conditions, interest rates, market volatility, and recession-sensitive signals. In addition to describing current levels, the dashboard incorporates regime comparisons, lead-lag diagnostics, inflation persistence measures, and yield-curve-based risk signals to assess whether present conditions are most consistent with expansion, late-cycle slowing, or rising recession risk.

Rather than relying on any single indicator, this framework emphasizes cross-validation across multiple dimensions of the macro environment. This is particularly important because macroeconomic relationships are often nonlinear, delayed, and unstable across policy regimes. As a result, the purpose of this report is not to make a deterministic forecast, but to provide a structured interpretation of current conditions and the balance of risks implied by the data

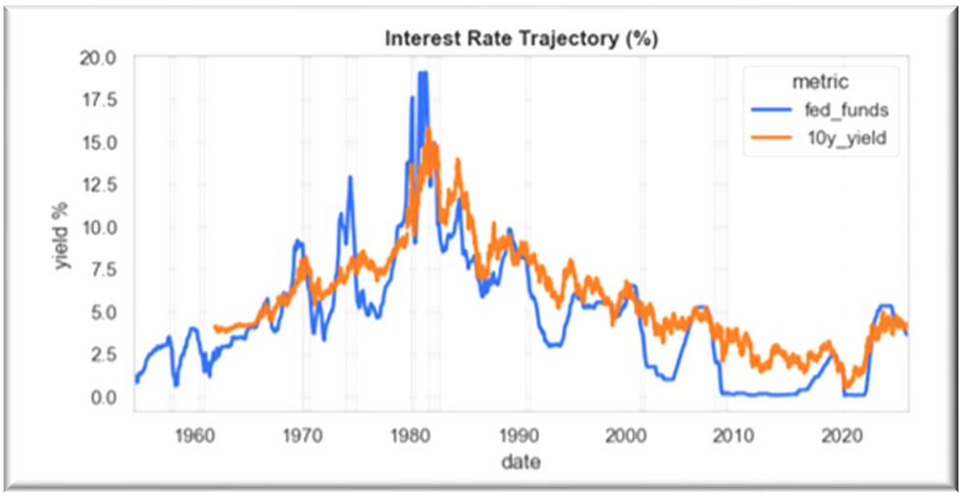
Broad Macro Environment

The overall dashboard suggests that the U.S. economy is not currently exhibiting conditions consistent with an active recession, though several indicators still support a cautious interpretation of the outlook. Historical percentile rankings indicate that real GDP and CPI remain elevated relative to their own long-run distributions, while unemployment remains comparatively low. At the same time, interest rates remain restrictive relative to the post-Global Financial Crisis era, and certain forward-looking indicators continue to imply that lagged downside risks have not fully dissipated.



From a business and investment perspective, the current environment appears most consistent with a late-cycle phase in which economic activity remains resilient, but the full effects of prior monetary tightening may still be unfolding.

Interest Rates and Policy Conditions



The interest rate trajectory chart shows that the federal funds rate remains elevated relative to the low-rate regime that characterized much of the period following 2008. Although current policy rates are not extreme when viewed against the full historical sample, they are meaningfully restrictive in a modern context. The 10-year Treasury yield has also moved materially higher from the ultra-low-rate environment of the 2010s, reflecting both the repricing of long-term inflation expectations and a normalization of the term structure of interest rates.

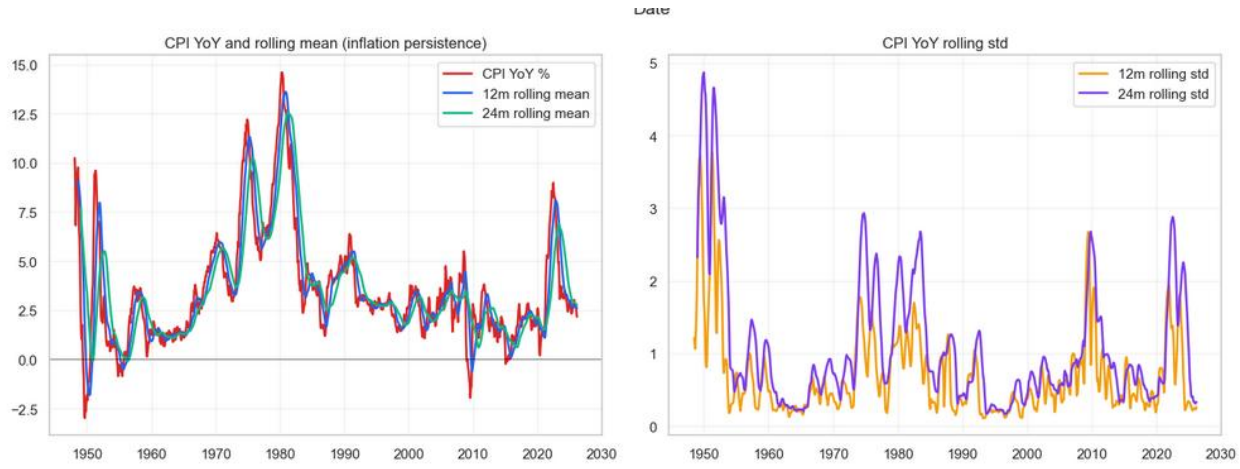
This combination suggests that financial conditions remain materially tighter than in the prior decade. Importantly, the divergence between the federal funds rate and the 10-year yield across different periods also highlights that longer-term yields are not determined by policy alone. They incorporate market expectations for future growth, inflation, and policy reversals. As a result, the interest rate backdrop should be interpreted not simply as “high rates,” but as a still-restrictive policy environment interacting with forward-looking market expectations.

The charts reinforce this interpretation. The historical level of the federal funds rate confirms that current policy is high relative to the post-crisis norm, while the as-of change charts show how sharply policy tightened during the recent cycle. For the policy rate specifically, percentage-change interpretations should be treated with caution because low starting values can mechanically create very large percentage moves. In practical terms, the substantive conclusion is that the policy stance tightened rapidly, not that the percent-change spikes themselves are economically meaningful.

For the 10-year yield, the level chart highlights the long secular decline from the 1980s through the 2010s, followed by a notable recent repricing upward. This indicates that long-term interest rates have normalized away from the zero-rate environment, but remain below the extremes observed in earlier decades.

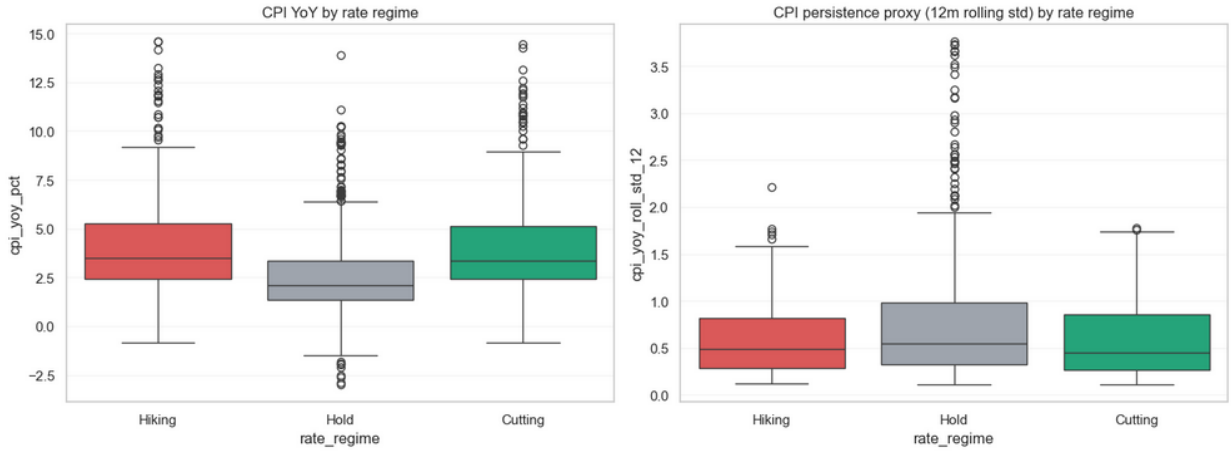
Inflation and Inflation Persistence

The inflation section of the dashboard indicates that inflation has **moderated significantly from its recent peak**, but has not yet fully returned to the stable, low-inflation regime that prevailed through much of the 2010s. The CPI annual percent change chart shows clear disinflation from the 2021–2022 surge, yet inflation remains above the lower range that markets and policymakers had become accustomed to in the pre-pandemic period.



The rolling mean and rolling standard deviation charts add an important second layer of interpretation. The rolling mean indicates that inflation shocks tend to persist rather than reverse immediately. Once inflation accelerates, it often remains elevated for a sustained period before gradually normalizing. The rolling standard deviation further shows that inflation has recently been not only higher than normal, but also more unstable. Although this instability has declined from peak levels, it has not fully disappeared.

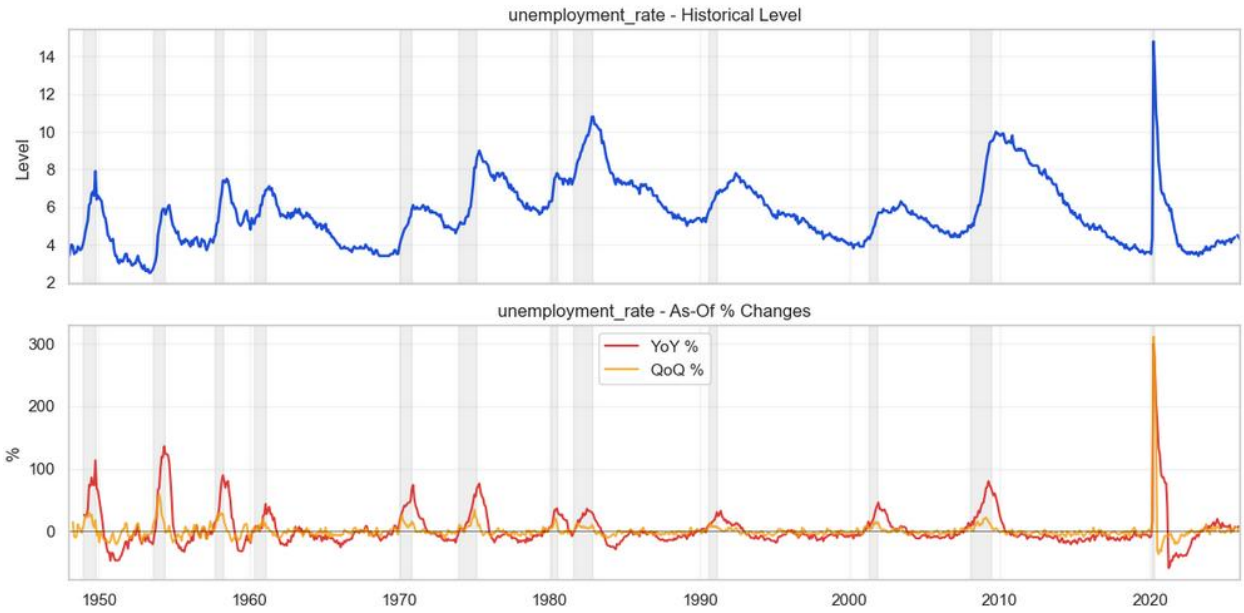
The rate-regime comparisons support this conclusion. Inflation tends to be lower and more stable during hold regimes, while hiking and cutting regimes are associated with both higher inflation and greater dispersion. This is a useful business interpretation: periods of active policy adjustment generally coincide with less predictable pricing conditions, which can affect margins, planning assumptions, and discount-rate expectations. Even when the central bank is not actively tightening, inflation persistence may remain present.



Taken together, the inflation evidence suggests that the inflation shock has cooled materially, but the process of full normalization remains incomplete. This supports a view of reduced but not eliminated inflation risk**.**

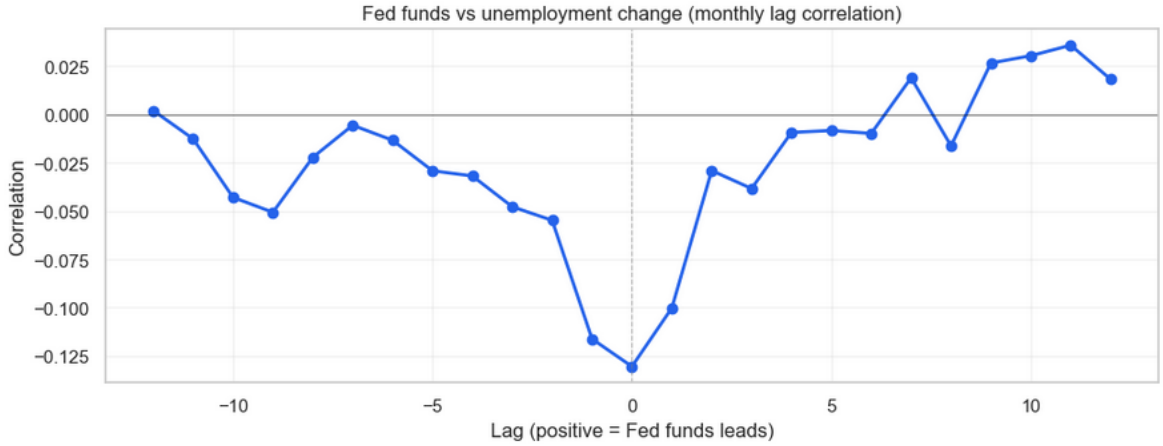
Labor Market Conditions

The labor market remains one of the strongest and most reassuring components of the dashboard. The unemployment rate remains low relative to its historical range, supporting the conclusion that the economy is not presently in a recessionary labor environment. Historically, recessions tend to coincide with sharp increases in unemployment, and the current level does not resemble those stress periods.

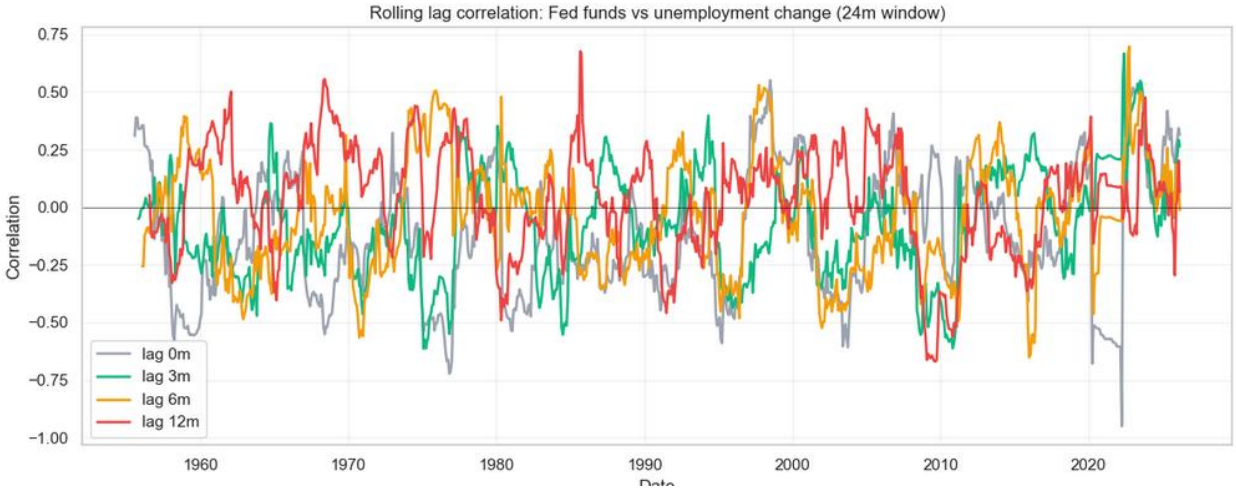


However, the unemployment charts also demonstrate that labor deterioration is often delayed and nonlinear. The unemployment rate can remain stable for an extended period and then increase rapidly once broader macro weakness begins to transmit into hiring and layoffs.

This is consistent with the lag-correlation results between federal funds changes and unemployment changes. The monthly lag profile suggests that policy effects on labor are not immediate, and in fact emerge only after several months. The rolling lag correlation makes the point even more clearly: the relationship is real, but it varies substantially over time.

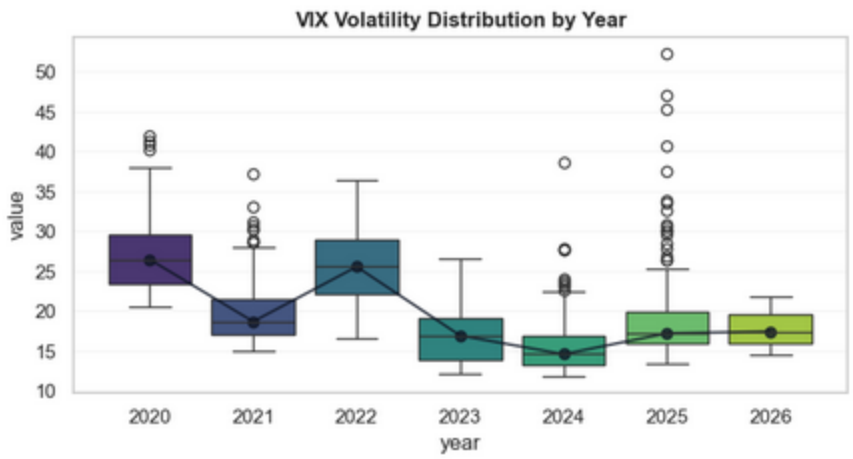


This has two important implications. First, a low unemployment rate is a strong indicator of current resilience. Second, it should not be interpreted as proof that macro risks have fully disappeared, because labor market deterioration often becomes visible only after policy tightening has already been in place for some time. For management and strategic planning purposes, the labor market should therefore be viewed as a **coincident-to-lagging strength indicator**, not a purely forward-looking one.

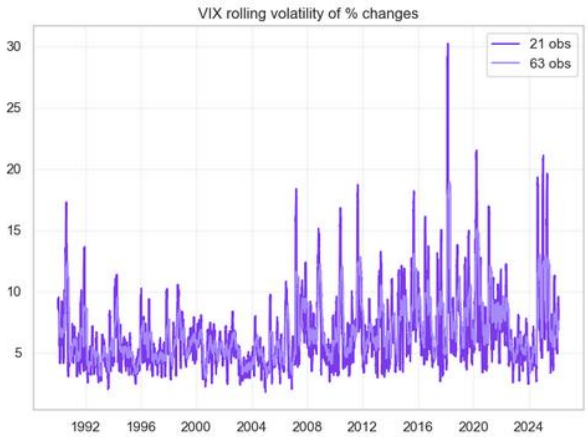


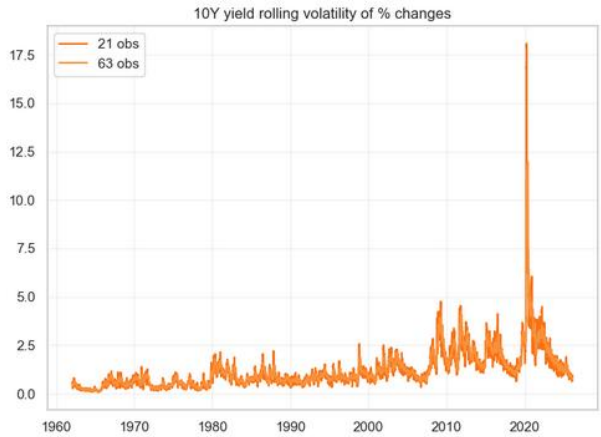
Market Volatility and Regime Comparisons

The VIX distribution and regime analysis add a market-based perspective to the macro picture. Recent VIX distributions show that baseline volatility has declined relative to peak-stress years, indicating that markets are pricing less acute systemic risk than during more turbulent periods. However, the continued presence of outliers shows that tail-risk episodes remain possible even in calmer volatility environments.



The regime-based event summary is particularly informative. Hiking, hold, and cutting periods produce different distributions for yield changes and volatility behavior. Hold regimes appear to be the most stable environment, with the lowest frequency of 10-year declines and VIX increases. By contrast, cutting regimes are associated with the highest probability that the 10-year yield falls, which is consistent with markets pricing weaker growth and lower forward rates. Hiking periods are more mixed: the 10-year still falls a meaningful share of the time, reflecting the market's tendency to anticipate that aggressive tightening may eventually restrain economic activity.





This section is especially important because it highlights that monetary easing should not automatically be interpreted as supportive for risk assets or macro sentiment. Cuts often occur in response to deteriorating conditions, and the associated market behavior may reflect that weakness rather than relief. In that sense, the regime analysis reinforces the broader conclusion that macro interpretation depends on **context and sequencing**, not just the direction of policy.

Relationships and Stability Across Time

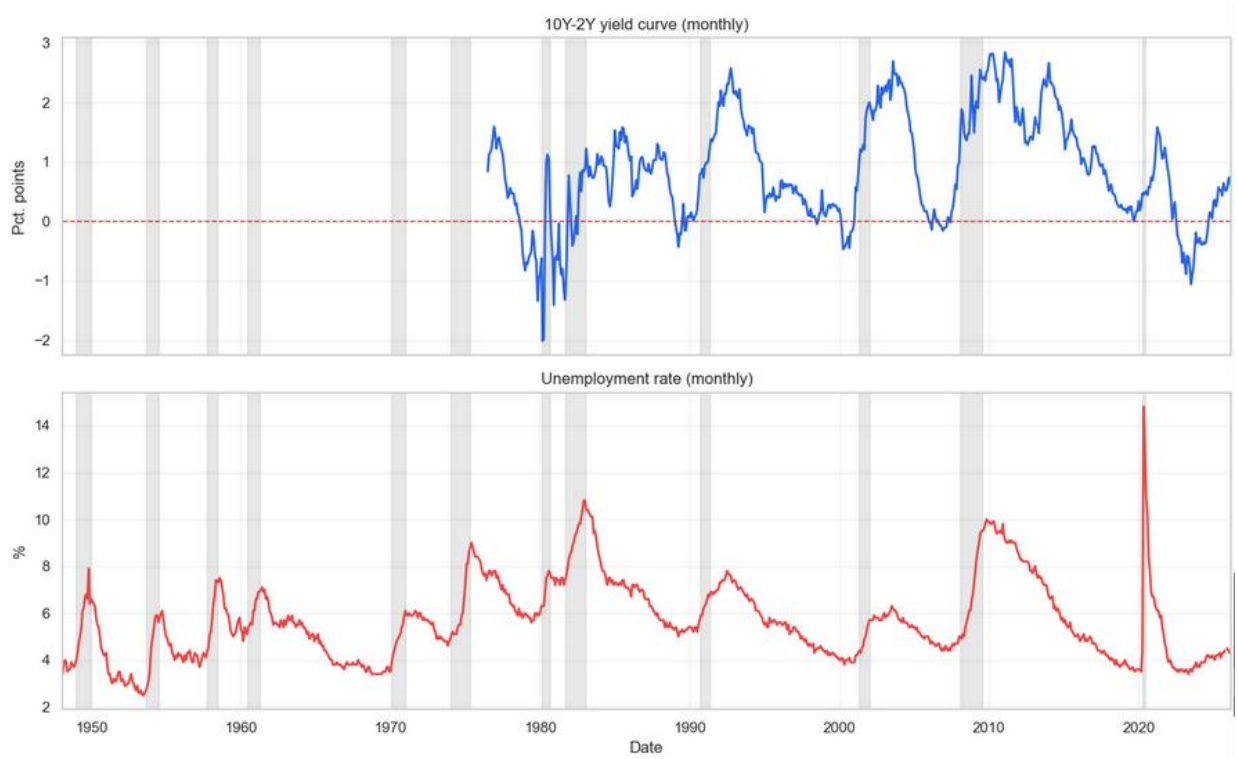
The VIX-versus-yield lead-lag analysis demonstrates that macro-financial relationships are not fixed across subperiods. The sign and timing of the relationship differ between the pre-2008, post-2008, and post-2020 samples. The strongest negative relationship appears in the post-2008 period, consistent with a classic risk-off pattern in which higher volatility coincides with falling yields. The post-2020 sample, however, is weaker and less stable in timing.

This instability is an important analytical result in its own right. It implies that these relationships should not be used as rigid forecasting rules. Instead, they are better interpreted as conditional diagnostics whose strength depends on the prevailing monetary regime, inflation environment, and market structure. For business and investment decision-making, this argues for a regime-aware framework rather than a static one.

A similar conclusion emerges from the policy-to-unemployment lag analysis. While tighter policy is associated with later labor market softness, the magnitude and timing of that relationship change meaningfully across rolling windows. This suggests that monetary transmission depends on the state of household balance sheets, corporate financing conditions, labor market tightness, and broader sentiment.

Yield Curve and Forward Risk Signals

The yield curve remains one of the most important forward-looking components of the report. The 10Y-2Y spread moved into inversion and has since turned positive again. Historically, inversion has often preceded periods of economic slowing or recession, while the subsequent re-steepening phase can occur closer to the point at which weakness becomes visible in the real economy.



The signal hit rates strengthen the interpretation of inversion as an early warning indicator. The probability that unemployment is higher 6 months after inversion is meaningful, and the 12-month unemployment probability is particularly strong. By contrast, the probability of recession within the next 12 months following inversion is materially lower. This suggests that inversion is more useful as a signal of future deterioration in labor and growth conditions than as a precise recession-timing mechanism.

In practical terms, the yield curve should therefore be treated as a medium-horizon risk signal rather than a literal countdown device. The fact that the curve has recently moved back above zero is a favorable

development relative to the inversion period, but it does not fully invalidate the information carried by the prior inversion. The more balanced interpretation is that immediate curve stress has eased, while lagged macro risk remains relevant**.**

Conclusion

The analysis presents a coherent and balanced view of the current macroeconomic environment. On one hand, several core indicators support the conclusion that the U.S. economy remains resilient. Real GDP remains strong relative to its historical distribution, unemployment remains low, and market volatility is well below peak-stress conditions. Inflation has also moved down substantially from its recent highs, reducing the immediacy of the earlier inflation shock.

On the other hand, the report does not support an "all clear" interpretation. Monetary policy remains restrictive relative to the modern era, inflation persistence has declined but not fully disappeared, and the lagged effects of earlier tightening may still emerge in labor and activity data. The lead-lag diagnostics show that these macro relationships are delayed and unstable rather than immediate and mechanical. The yield curve evidence further suggests that prior inversion still carries cautionary information, particularly with respect to slower growth and future labor market softening.

Overall, the most reasonable conclusion is that the economy is currently in a late-cycle, slowing-but-still-resilient phase. Present conditions do not indicate an active recession, but neither do they justify complacency. The balance of evidence points toward a macro environment in which resilience remains visible in current data, while forward-looking and lagged indicators continue to argue for disciplined monitoring of downside risk.

References

Board of Governors of the Federal Reserve System (US). (2026). *Federal Funds Effective Rate (FEDFUNDS)*. Retrieved Feb 12, 2026, from FRED, Federal Reserve Bank of St. Louis: https://fred.stlouisfed.org/series/FEDFUNDS

Chicago Board Options Exchange. (2026). *CBOE Volatility Index: VIX (VIXCLS)*. Retrieved Feb 12, 2026, from FRED, Federal Reserve Bank of St. Louis: https://fred.stlouisfed.org/series/VIXCLS

Federal Reserve Bank of St. Louis. (2026). *10-Year Treasury Constant Maturity Minus 2-Year Treasury Constant Maturity (T10Y2Y)*. Retrieved Feb 12, 2026, from FRED, Federal Reserve Bank of St. Louis: https://fred.stlouisfed.org/series/T10Y2Y

Federal Reserve Bank of St. Louis. (2026). *Unemployment Rate (UNRATE)*. Retrieved Feb 12, 2026, from FRED, Federal Reserve Bank of St. Louis: https://fred.stlouisfed.org/series/UNRATE

U.S. Bureau of Economic Analysis. (2026). *Gross Domestic Product (GDP)*. Retrieved Feb 12, 2026, from FRED, Federal Reserve Bank of St. Louis: https://fred.stlouisfed.org/series/GDP

U.S. Bureau of Labor Statistics. (2026). *Consumer Price Index for All Urban Consumers: All Items in U.S. City Average (CPIAUCSL)*. Retrieved Feb 12, from FRED, Federal Reserve Bank of St. Louis: https://fred.stlouisfed.org/series/CPIAUCSL

U.S. Bureau of Labor Statistics. (2026). *Unemployment Rate (UNRATE)*. Retrieved Feb 12, from FRED, Federal Reserve Bank of St. Louis: https://fred.stlouisfed.org/series/UNRATE

Weinstock, L. R. (2024, September 30). *The Debt Limit* (CRS In Focus No. IF12774). Congressional Research Service. https://crsreports.congress.gov/product/pdf/IF/IF12774

Appendix

Table 1:

Start Date	End Date	Duration (months)
2020-03	2020-04	2
2008-01	2009-06	18
2001-04	2001-11	8
1990-08	1991-03	8
1981-08	1982-11	16
1980-02	1980-07	6
1973-12	1975-03	16
1970-01	1970-11	11
1960-05	1961-02	10
1957-09	1958-04	8
1953-08	1954-05	10
1948-12	1949-10	11
1945-03	1945-10	8

1937-06	1938-06	13
1929-09	1933-03	43
1926-11	1927-11	13
1923-06	1924-07	14
1920-02	1921-07	18
1918-09	1919-03	7
1913-02	1914-12	23
1910-02	1912-01	24
1907-06	1908-06	13
1902-10	1904-08	23
1899-07	1900-12	18
1896-01	1897-06	18
1893-02	1894-06	17
1890-08	1891-05	10
1887-04	1888-04	13

1882-04	1885-05	38
1873-11	1879-03	65
1869-07	1870-12	18
1865-05	1867-12	32
1860-11	1861-06	8
1857-07	1858-12	18
1854-12	1854-12	1